Class 10 NCERT Maths
CLASS 10 NCERT MATHS
Chapter 8: Introduction to Trigonometry
CHAPTER 8: INTRODUCTION TO TRIGONOMETRY

A simplified, detailed guide in Hinglish
For students who missed Class 7-8-9
Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

1.1 TRIGONOMETRY KYA HAI?

"Tri-gon-metry" = teen (tri) + kona/angle (gon) + measurement (metry).

Matlab: TRIANGLES ke angles aur sides ka rishta study karna - khaaskar RIGHT-ANGLED triangle.

1.2 RIGHT-ANGLED TRIANGLE KE PARTS

Ek angle theta (let's call it) ke reference se:

```
           C
           |\
           | \
PERPEN-    |  \   HYPOTENUSE
DICULAR    |   \  (always opposite to 90 deg)
(opposite  |    \
 to theta) |     \  theta
           |______\
B  BASE  A
           (adjacent to theta)
```

3 sides:
- HYPOTENUSE: 90 degree ke saamne wali (longest)
- PERPENDICULAR (Opposite): theta ke saamne wali
- BASE (Adjacent): theta ke saath wali (na hyp)

1.3 RATIO YAAD KARNE KA TRICK

"Pandit Badri Prasad
 Har Har Bole

Sona Chandi Tole"

P / H = Perpendicular / Hypotenuse = sin
B / H = Base / Hypotenuse = cos
P / B = Perpendicular / Base = tan

(Pehla letter pair upar/neeche dekh)

SECTION 2: SIX TRIGONOMETRIC RATIOS

2.1 THE THREE MAIN RATIOS

For angle theta in a right triangle:

sin(theta) = Perpendicular / Hypotenuse = P/H
cos(theta) = Base / Hypotenuse = B/H
tan(theta) = Perpendicular / Base = P/B

2.2 THE THREE RECIPROCAL RATIOS

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cosec(theta) = 1/sin = H/P
sec(theta) = 1/cos = H/B
cot(theta) = 1/tan = B/P

2.3 USEFUL RELATIONS

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tan(theta) = sin(theta) / cos(theta)
cot(theta) = cos(theta) / sin(theta)

sin x cosec = 1
cos x sec = 1
tan x cot = 1

2.4 EXAMPLE

EXAMPLE 1: Right triangle: Perpendicular=3, Base=4.
Hypotenuse aur ratios nikaal.

Hyp = sqrt(3^2 + 4^2) = sqrt(25) = 5 (3-4-5 triplet)

sin = 3/5, cos = 4/5, tan = 3/4
cosec = 5/3, sec = 5/4, cot = 4/3

SECTION 3: SPECIFIC ANGLES (YAAD RAKH!)

3.1 THE MASTER TABLE

Angle:	0	30	45	60	90
sin	0	1/2	1/sq2	sq3/2	1
cos	1	sq3/2	1/sq2	1/2	0
tan	0	1/sq3	1	sq3	undef

(sq2 = sqrt2 = 1.414, sq3 = sqrt3 = 1.732)

3.2 YAAD KARNE KA EASY TRICK (sin ke liye)

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sin values: write 0,1,2,3,4 -> divide by 4 -> sqrt

Angle:	0	30	45	60	90
Step1:	0	1	2	3	4
/4:	**0/4**	**1/4**	**2/4**	**3/4**	**4/4**

sqrt: sqrt0 sqrt(1/4) ...

= 0, 1/2, 1/sqrt2, sqrt3/2, 1

cos = same table ULTA (reverse order)
tan = sin / cos

3.3 EXAMPLES

EXAMPLE 2: sin 30 + cos 60 = ?
= 1/2 + 1/2 = 1

EXAMPLE 3: tan 45 + cos 0 = ?
= 1 + 1 = 2

EXAMPLE 4: 2 sin 30 x cos 60 = ?
= 2 x (1/2) x (1/2) = 1/2

SECTION 4: TRIGONOMETRIC IDENTITIES

4.1 THE THREE IDENTITIES (BAHUT IMPORTANT)

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1. $\sin^2(\theta) + \cos^2(\theta) = 1$

2. $1 + \tan^2(\theta) = \sec^2(\theta)$

3. $1 + \cot^2(\theta) = \text{cosec}^2(\theta)$

In se aur bhi forms nikalte hain:

$\sin^2 = 1 - \cos^2$
$\cos^2 = 1 - \sin^2$
$\sec^2 - \tan^2 = 1$
$\text{cosec}^2 - \cot^2 = 1$

4.2 EXAMPLES

EXAMPLE 5: Agar $\sin(\theta) = 3/5$, $\cos(\theta)$ nikaal.

$\sin^2 + \cos^2 = 1$
$(3/5)^2 + \cos^2 = 1$
$9/25 + \cos^2 = 1$
$\cos^2 = 16/25$
$\cos = 4/5$

EXAMPLE 6: Prove: $(1 - \cos^2 A) \times \text{cosec}^2 A = 1$

LHS $= (1 - \cos^2 A) \times \text{cosec}^2 A$
$= \sin^2 A \times \text{cosec}^2 A$ $[1-\cos^2 = \sin^2]$
$= \sin^2 A \times (1/\sin^2 A)$
$= 1 =$ RHS   (Proved)

Q AND A TIME

Q1. Right triangle: Perpendicular = 5, Base = 12.
Hypotenuse aur sin, cos, tan nikaal.

Q2. Value nikaal:
(a) $\sin 60 + \cos 30$
(b) $\tan 30 \times \tan 60$
(c) $\cos 0 - \sin 90$

Q3. Agar $\cos(\theta) = 12/13$, toh $\sin(\theta)$ nikaal
(identity use kar).

Q4. Prove: $\sin^2 30 + \cos^2 30 = 1$

(values daal ke verify kar)

Q5. sec^2 45 - tan^2 45 = ? (identity ya values se)

SUMMARY

1. 6 ratios:
   sin = P/H, cos = B/H, tan = P/B
   cosec = H/P, sec = H/B, cot = B/P

2. Trick: "Pandit Badri Prasad..." for sin, cos, tan.

3. Special angles table (0,30,45,60,90) yaad rakh.

4. Three identities:
   sin^2 + cos^2 = 1
   1 + tan^2 = sec^2
   1 + cot^2 = cosec^2

5. tan = sin/cos, reciprocals: sin-cosec, cos-sec, tan-cot.

TIPS FOR EXAM

1. Special angle table HARD yaad kar - har question mein use hoti hai.

2. sin trick (0,1,2,3,4 / 4 -> sqrt) se table jaldi bana sakta hai agar bhool jaye.

3. Identity proofs mein hamesha COMPLEX side se shuru karke simple side tak pahuncho.

4. sin^2 ka matlab (sin theta)^2 hota hai, sin(theta^2) nahi - dhyaan rakh.

5. tan 90 aur sec 90 UNDEFINED hote hain (denominator 0).

Generated by Kiro for Krishna1k
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